LESSON PLAN

EMI/EMC (14)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	15
Total No. of Lesson Plans	46	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Need and role of flight control system, fundamental elements of flight control system
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Need and role of flight control system, fundamental elements of flight control system.
5. Specific Objectives	14.2.1.1 Trainee will be able to explain the functions of AFCS.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of AFCS functional diagram; Instructor supervision required for diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the three primary axes of aircraft rotation?
(b) Why is manual control insufficient for high-speed flight?
(c) Define the term 'stability' in the context of flight dynamics.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition and necessity of flight control systems • Role of AFCS in modern aircraft operations • Distinction between primary and secondary controls • Impact of pilot workload on manual control • Basic classification of control surfaces	L & Ds	PPT, IB	15 min
(b)	• Fundamental elements: Control stick and rudder pedals • Fundamental elements: Control surfaces and hinges • Fundamental elements: Control linkages and cables • Fundamental elements: Hydraulic actuators and servos • Fundamental elements: Feedback mechanisms and feel units	L & Ds	PPT, PDF	20 min
(c)	• Integration of AFCS with autopilot systems • Safety considerations in flight control design • Maintenance implications of control system elements • Troubleshooting common control system failures • Future trends in fly-by-wire architectures	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the AFCS reduces pilot workload during turbulence.
(b) Identify the fundamental element responsible for converting pilot input into surface movement.
(c) Describe the role of feedback mechanisms in ensuring control feel.

12. Home Assignment and Follow Up	Prepare a brief note on how the principles of flight control systems discussed in this lesson apply to the functional diagram of a basic closed loop flight control system covered in LP 16.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Functional diagram of a basic closed loop flight control system. Explanation of automatic control and classification of system and explanation of controls.	Refer to the master syllabus reference list	LP 16

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min